UNIT-6

Choose the correct answer:

1. c) both covalent crystals
2. b) AB_3

   number of A ions $= \left(\frac{N_c}{8}\right) = \left(\frac{8}{8}\right) = 1$

   number of B ions $= \left(\frac{N_f}{2}\right) = \left(\frac{6}{2}\right) = 3$

   simplest formula AB_3
3. b) 1:2

   if number of close packed atoms =N; then,

   The number of Tetrahedral holes formed = 2N

   number of Octahedral holes formed = N

   therefore N:2N = 1:2
4. c) molecular solid

   lattice points are occupied by CO_2 molecules
5. a) Both assertion and reason are true and reason is the correct explanation of assertion.
6. c) 8 and 4

CaF_2 has cubical close packed arrangement

Ca^{2+} ions are in face centered cubic arrangement, each Ca^{2+} ions is surrounded by 8 F^- ions and each F^- ion is surrounded by 4 Ca^{2+} ions.

Therefore coordination number of Ca^{2+} is 8 and of F^- is 4

7. b) 6.023×10^{22}

in BCC unit cell,

2 atoms $\equiv$ 1 unit cell

Number of atoms in 8g of element is ,

$$\text{Number of moles} = \frac{8g}{40 \text{ g mol}^{-1}} = 0.2 \text{ mol}$$

1 mole contains 6.023×10^{23} atoms

0.2 mole contains $0.2 \times 6.023 \times 10^{23}$ atoms

$$\left(\frac{1\text{unit cell}}{2 \text{ atoms}}\right) \times 0.2 \times 6.023 \times 10^{23}$$

6.023×10^{22} unit cells

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